



DPUSGEView

Represents an output window and a camera in SILAB's visualization SGE.

1. Inputs:

<i>X</i>	X coordinate of the vehicle position
<i>Y</i>	Y coordinate of the vehicle position
<i>Z</i>	Z coordinate of the vehicle position
<i>yaw</i>	Yaw angle of the vehicle position (rotation about the Z axis, radians)
<i>pitch</i>	Pitch angle of the vehicle position (rotation about the Y axis, radians)
<i>roll</i>	Roll angle of the vehicle position (rotation about the X axis, radians)
<i>pitchRoad</i>	Pitch angle of the road at the camera's position
<i>rollRoad</i>	Roll angle of the road at the camera's position

2. Dynamic parameters:

<i>CamA</i>	Height (m) of the driver's head above the vehicle position. Default: 0.8 m
<i>CamB</i>	Horizontal distance (m) of the driver's head to the vehicle position. Default: 0.6 m
<i>CamBeta</i>	Direction of the CamB vector (degrees). Default: -90° (to the left)
<i>CamS</i>	Height (m) of the vehicle's center of gravity. Default: 0.34 m
<i>Mirror</i>	Boolean flag which mirrors the horizontal direction (for use in rear mirror views). Default: 0
<i>CamRoadZOffset</i>	Difference of the height (m) of the road at the point the driver is (probably) looking at, to the road height below the vehicle (Z axis). Default: 0
<i>LodDistanceMul</i>	Multiplier for the level-of-detail distances specified in the loaded graphical objects. The higher the number is, the further away the most detailed version of each object is drawn. However, this may also reduce the frame rate. Default: 1
<i>ProjectionOffsetX</i> , <i>ProjectionOffsetY</i>	Offset of the center of the perspective projection (the vanishing point) from the center of the view window. The default values of 0 put the vanishing point in the middle of the view. Positive values (up to +1) are to the right and to the top, negative values (up to -1) are to the left and to the bottom.

3. Dynamic parameters for use with monitors or projectors only:

<i>CamGamma</i>	View angle (degrees) of the camera (rotation about the Z axis vs. the vehicle orientation). Default: 0°
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CamDelta Downtilt (degrees) of the camera (rotation about the Y axis vs. the vehicle orientation). Default: 0°

4. Dynamic parameters for use with head-mounted displays only:

HeadYaw Yaw angle of the driver's head (rotation about the Z axis, radians). Default: 0

HeadPitch Pitch angle of the vehicle position (rotation about the Y axis, radians). Default: 0

HeadRoll Roll angle of the vehicle position (rotation about the X axis, radians). Default: 0

EyeX Translation of the eye (camera) position from the head's middle axis along the X axis. Default: 0 m

EyeY Translation of the eye (camera) position from the head's middle axis along the Y axis. Default: 0 m

EyeZ Translation of the eye (camera) position from the head's middle axis along the Z axis. Default: 0 m

OverlayDistance Virtual distance of the overlay objects from the camera (in meters). This parameter affects stereo views only, where $EyeY \neq 0$. Default: ∞ , recommended value: about 2 (a bit less than the distance to the road).

5. Static parameters:

CamFOVY Camera's vertical field of view (degrees). Default: 47°

CamNearClipping Distance of the nearest visible object (m). Default: 1.1, must be at least 0.1.
Warning: Values less than 1 can lead to visual problems (e.g. incorrectly displayed trees) and should be avoided if possible.

CamFarClipping Distance of the most distant visible object (m). Default: 128000
Warning: Values less than 35000 will lead to incorrect rendering of the panorama and should not be used. The value should also not be increased above 150000.

VisTest Boolean flag which indicates if the visibility test is used for this view. It doesn't render objects that can't be seen from the current camera perspective and improves the frame rate by 20-30% in many situations. Default: *true*

Mask Allows masking part of the view, i.e. this part will always be black. More information about masks is available in the corresponding section on page 8.

6. Static parameters which configure the viewing window:

WindowX X coordinate of the viewing window's top-left edge (pixels)

WindowY Y coordinate of the viewing window's top-left edge (pixels)

WindowH Height of the viewing window (pixels). Default: 1024

WindowW Width of the viewing window (pixels). Default: 768

WindowAspectRatio Aspect ratio of the viewing window, compared to $WindowH/WindowW$. Default: 1.0



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<i>WindowShift</i>	Vertical displacement of the image (pixels). Deprecated: Please use the <i>ProjectionOffsetY</i> parameter instead.
<i>WindowHShift</i>	Horizontal displacement of the image (pixels). Deprecated: Please use the <i>ProjectionOffsetX</i> parameter instead.
<i>WindowTopmost</i>	If set to true, the viewing window is held “always on top” (like the Windows taskbar). Default: <i>false</i>
<i>ShareView</i>	If set to the DPU instance name of an already existing view (whose <i>DPU Index</i> is lower), this view is laid over the other view’s window. In this case, the parameters <i>WindowXYWH</i> are ignored, and the parameters <i>ViewportXYWH</i> are used instead. Note: When specifying a shared view, all Viewport parameters have to be explicitly defined even if they are set to 0. Default: (<i>none</i>), which means this is an independent view
<i>AllowSharing</i>	Allows sharing some OpenGL resources with a second <i>SGEView</i> . Possible values are: 15 (default): All suitable resources are shared. This leads to the fastest update rate of the visualization. 1: Shadow buffer is not shared. Use this setting if the shadows are not displayed correctly (e.g. if the camera positions are vastly different) 0: Textures are not shared. Use this setting only if the view is rendered using a second graphics card.
<i>ViewportX, ViewportY</i>	Position of the view’s top-left edge (pixels), relative to the containing window
<i>ViewportH, ViewportW</i>	Size of the viewing area (pixels)

7. Static parameters concerning rendering:

<i>EnableTextureAnisotropy</i>	Use anisotropic filtering for textures? Improves the display quality, especially of the ground and of houses, but may reduce the frame rate. Default: <i>true</i>
<i>LowTextureAnisotropy</i>	Amount of anisotropic filtering for “unimportant objects” Default: 2, usually: 0..16
<i>MedTextureAnisotropy</i>	Amount of anisotropic filtering for “somewhat important objects” Default: 4, usually: 2..16
<i>MaxTextureAnisotropy</i>	Amount of anisotropic filtering for “very important objects” Default: 16, usually: 2..16
<i>AntiAliasTransparency</i>	Boolean flag which smoothens borders of irregularly shaped objects (e.g. trees). Default: <i>true</i> The effect has little influence on frame rate, but only increases the display quality on newer graphics boards (since GeForce Series 8) and only works when the parameter <i>AntiAlias</i> of <i>DPUSGEWorld</i> is set to more than 1.
<i>UnloadTextures</i>	Unload the textures from main memory after sending them to the graphics card? Default: <i>false</i>



This can make it possible to launch large scenarios on limited machines, but may prevent launching the simulation for more than one time and should only be used if needed.

8. Static parameters concerning flow control:

<i>Threaded</i>	(0..5) Number of the processing thread in which this SGEView runs. 0 is the main thread, in which most other DPUs run. Default: 0 It is recommended to set <i>Threaded=1</i> if at least a dual-core processor is available.
<i>VerticalSync</i>	Boolean flag. If set to true and the view is running in its own thread (<i>Threaded</i> $\neq$ 0), screen refresh is synchronized with the monitor. This increases display quality, especially while turning. Default: <i>false</i>

9. Static parameters and inputs for screenshot capturing:

<i>ScreenshotDir</i>	Specifies the folder in which screenshot files are saved. If the path isn't absolute, creates the folder below SILAB's install folder. Default: (<i>empty</i>), e.g. corresponding to D:\SILAB
<i>ScreenshotBaseName</i>	Base name for screenshots. A serial number is automatically appended to the base name. Default: <i>Screenshot</i>
<i>ScreenshotExtension</i>	Filename extension for screenshots. This also defines the image format that is used for the screenshots. Default: <i>tga</i> Warning: If an extension other than <i>tga</i> is used, screenshots are always written at once (as if <i>WriteAtOnce=true</i> was set). This will lead to time lags and stuttering on the computer where the screenshots are made.
<i>WriteAtOnce</i>	Boolean flag which indicates if a screenshot is saved at once. As this may lead to time lags and stuttering on the computer, the default value is false, meaning that the screenshot save is divided into several simulation steps.
<i>WritePerStep</i>	If <i>WriteAtOnce=false</i> , the number of bytes written per simulation step. Default: 100000
<i>Screenshot</i>	When switching from 0 to 1, a screenshot is captured. Before the next screenshot can be created, the variable must be reset to 0. If the screenshot is not written at once (<i>WriteAtOnce=false</i>), a new screenshot cannot be created before the old screenshot has been written to disk completely.
<i>ScreenshotOffscreen</i>	If activated (<i>true = default value</i>), the screenshot is, if possible, captured from an off-screen part of the graphics memory. This is only possible if the nighttime view is enabled (<i>SGEWorld.RenderMode</i> = 5 or 6) or if a <i>SGEDistortion</i> DPU has been created for this view. Off-screen captures differ in the following aspects: - No distortion is visible in the screenshot - The screenshot is captured correctly even if the viewing window is obscured by other windows



ScreenshotTransparency

- Screenshots can be captured at higher resolutions than what the connected monitor can display. (See section “High-resolution screenshots”).)

If set to *true*, screenshots contain transparency information (i.e. the image file contains an alpha channel). The default value of false creates completely opaque screenshots.

10. Outputs:

FPS

Current frame rate, which is the number of rendered images within the last second.

11. Dependencies to other DPUs:

DPUSGEView must run together with **DPUSGEWorld**. The Index value of **DPUSGEView** must be higher than **DPUSGEWorld**'s Index value.

12. Functional principle:

An instance of **DPUSGEView** is required on each computer which displays a view of the SGE 3D world. It manages the visualization's output window and the camera position, which can be selected as desired.

A SGE 3D world is a collection of three-dimensional graphics objects which are administered by **DPUSGEWorld** and displayed by **DPUSGEView**. The objects are inserted and animated by individual visualization DPUs, such as **DPUSCNXSGE** and **DPUTRFSGE**.

13. Multiple DPUSGEView instances:

It is possible to display more than one view of the SGE 3D world on one computer. The following limits apply:

- With a workstation license, up to two instances of **DPUSGEView** can be created.
- With a simulator license, up to six instances of **DPUSGEView** can be created on each connected computer.

In order to use the graphics hardware efficiently and achieve the highest possible frame rate, the following points should be considered when displaying more than one instance of **DPUSGEView**: (Let us assume that the instances are named **SGEView**, **SGEView2** up to **SGEViewN**).

- All **DPUSGEView** instances should be created in a chain with ascending Index values: **SGEWorld.Index** < **SGEView.Index** < **SGEView2.Index** < ... < **SGEViewN.Index**.
- For all **DPUSGEView** instances except for the first one (**SGEView2** to **SGEViewN**) **ViewportXYWH** must be set explicitly defined if they are set to 0.
- Each **SGEView** instance should share the data of the previous instance in the chain:

```
SGEView.ShareView = "";  
SGEView2.ShareView = "SGEView";  
SGEViewN.ShareView = "SGEViewN-1";
```

- All views should be displayed in the same window (which may possibly span several screens), and be positioned within the window using the *Viewport* parameters.
- All DPUSGEView instances should be created within the same thread (*SGEViewi.Threaded* = 1 for all of the views).
- The *VerticalSync* option should be disabled for all DPUSGEView instances except the first one.

```
SGEView.VerticalSync = true; [...]  
SGEViewN-1.VerticalSync = false;  
SGEViewN.VerticalSync = false;
```

14. Positioning the Camera:

The camera position is determined using a three-step process. This process depends on the visualization device.

For fixed monitors or projectors, the process is as follows:

- The vehicle position is specified using the inputs X, Y, Z and its current orientation is specified using *yaw*, *pitch*, *roll*, *pitchRoad*, *rollRoad*.
- Within the moving vehicle, the parameters *CamS*, *CamA*, *CamBeta* and *CamB* are used to specify the position of the driver's head.
- *CamGamma* and *CamDelta* are used to specify the orientation of the monitor. The resulting position is used for the scene camera (see Figure 1).

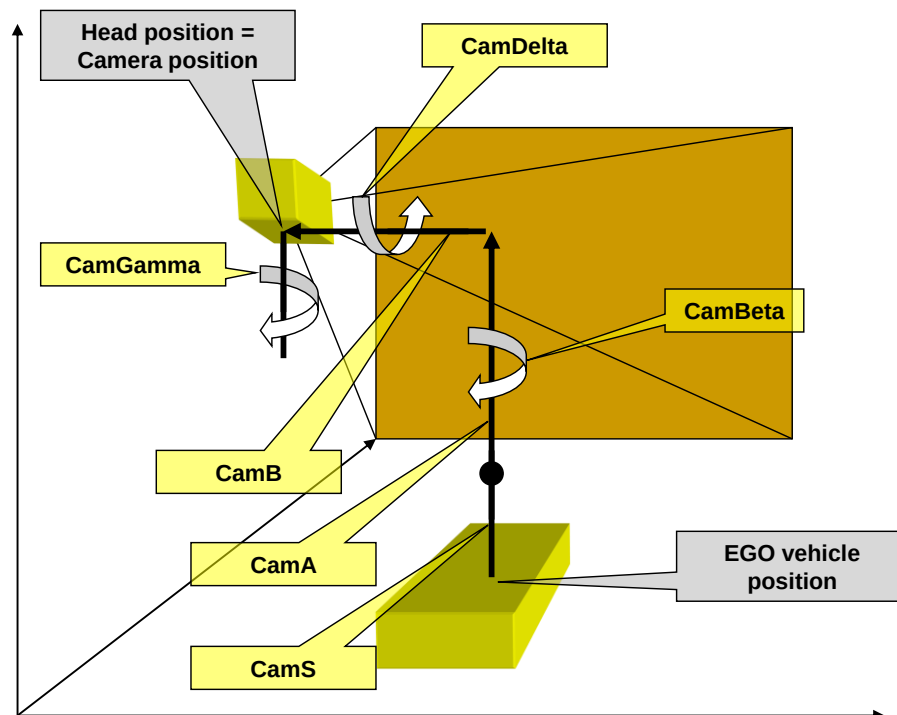


Figure 1: Camera positioning for monitors and projectors



16. Masks:

In some situations, it is necessary to always display some sections of a view in black. This is especially necessary when a projector illuminates parts of the simulator mockup. The parameter Mask can be used for that purpose. A number of tuples is specified for this parameter. Each of the tuples represents a convex polygon that will be cut out from the view. Within the polygon, the edges are specified in counter-clockwise order and X- and Y- coordinates are specified alternatingly.

Example:

```
DPUSGEView View {  
    ...  
    WindowW = 1024;  
    WindowH = 768;  
    Mask = {(1024, 768, 1024, 568, 824, 768)};  
};
```

This defines a mask which consists of a single triangle, and cuts away the bottom-right corner of the viewing window:



By adding a second triangle, the bottom-left edge is also cut out:

```
DPUSGEView View {  
    ...  
    Mask = {(1024, 768, 1024, 568, 824, 768),  
            (0, 768, 200, 768, 0, 568)};
```




```
};
```

If the simulator contains more than one graphics view (for example, three projectors and three mirror displays), the mask is usually defined separately for each of the views. To do so, the parameter Mask is specified outside of the DPU instantiation, with prepended computer and DPU name:

```
FRONT.View.Mask = {(1024, 768, 1024, 568, 824, 768)};
```

Either the actual computer name or alias names (such as *VISF01*) can be used here.

17. High-resolution Screenshots:

When printing (e.g. posters), screenshots with a very large resolution are often desired. However, the screenshot resolution is normally limited by the display resolution. Since SILAB 3.0, this limitation can be circumvented by the *ScreenshotOffscreen* function of *DPUSGEGView*.

The *ScreenshotOffscreen* function performs the screenshot in an invisible part of the graphics memory, which can be larger than the monitor's resolution. To activate off-screen rendering, the "distortion" pipeline must be activated. This can be achieved by adding *DPUSGEDistortionDummy* to the configuration.

The following DPU configuration creates – on a monitor with only 1024x768 pixels – screenshots with quadruple resolution.

```
Pool Screenshot : Full
{
    Executable = true;

    DPUSGEDistortionDummy SGEDistortion
    {
        Computer = {VIS};
        Index = 50;

        # Select screenshot resolution
        ResolutionX = 4096;
        ResolutionY = 3072;
    };

    SGEGView.WindowW = 1024;
    SGEGView.WindowH = 768;
    SGEGView.ScreenshotOffscreen = true;
};
```

Note: Depending on the graphics board, there are still limitations when using this method. When these are exceeded, the following message is shown when preparing the simulation:

(!) cSGERenderBuffer : CheckFrameBufferStatus : Unsupported framebuffer format



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The display is then incorrect. Current NVidia graphics boards (e.g. GeForce GTS 250 and GTX 460) allow maximum resolutions of 8 megapixels. If a 4:3 format is used, this leads to a maximum resolution of 3200 x 2400 pixels.